Tips for Delivering an Effective Lab Report

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***Abstract* — We present some guidelines to follow in order to prepare well your labs. The main programming language you are going to use to implement some NLP applications is Python. The codes you will develop run on top of a Docker image, ensuring a consistent and reproducible environment. You will use the interactive tool Marimo. It is preferable to write your lab reports in Typst, given the provided files.**

***Index terms* — Python, Typst, GitHub, Docker, Lab Report**

I. Python

The programming language we are going to learn through this module is **Python**. Figure 1 shows its logo. It is chosen for:

- its high-prototyping capabilities,
- expressive syntax, and
- extensive ecosystem.

Figure 1: Python logo

Insert your code in each lab report. The detailed explanation of the functions, along the packages you have used is a must. After each code snippet, add a screenshot that showcases your running work when applying the test provided in each exercise.

II. Typst

Consider using **Typst** to write your lab reports. The provided templates allow you to focus on the content and seamlessly create a professional-looking report.

Typst supports Markdown syntax, which provides a range of formatting options [1]. Here are some points to help you write your report:

1. Formatting Text:
   - Surround a text with single asterisks '(*)' to make it bold
   - Use single underscores '(_)' around your text to emphasize it
   - To create headings, use equal signs '(=)' followed by a space at the begining of a line. The number of '(=)' symbols determines the heading level.
2. Creating lists:
   - Unordered list: use a hyphen '(-)' followed by a space for each list item
   - Ordered list: use a plus sign '(+)' followed by a space for each list item
3. Code snippets:
   - Inline code: enclose the code within backticks (`)
   - Block of code: use triple backticks followed by the word 'julia' to enable syntax highlighting

```python
import spacy
spacy.load('en_core_web_sm')
```

4. Inserting Objects:
   - Use this syntax if you need to insert an image:

```
#figure(
  image("IMAGE_NAME.EXT", width:
100%),
  caption: [IMAGE_CAPTION],
) <fig:LABEL>

@fig:LABEL shows an image.
```

   - Use this syntax if you need to draw a table:

```
#figure(
  table(
    columns: 4,
    [Row 1], [a], [b], [c],
```

```
    [Row 2], [1], [2], [3],
  ),
  caption: [Results],
) <tab:LABEL>

@tab:LABEL displays some results.
```

In each document, you have to insert well annotated screenshots of your code after being executed.

You can leverage those features using the app's intuitive interface at the url typst.app, as shown in Figure 2. No installation is required, however, you may need to sign in in order to use the online editor. Keep an eye on your project size. Do not exceed 200MB. A fully fledged documentation on the usage of **Typst** is available at typst.app/docs.

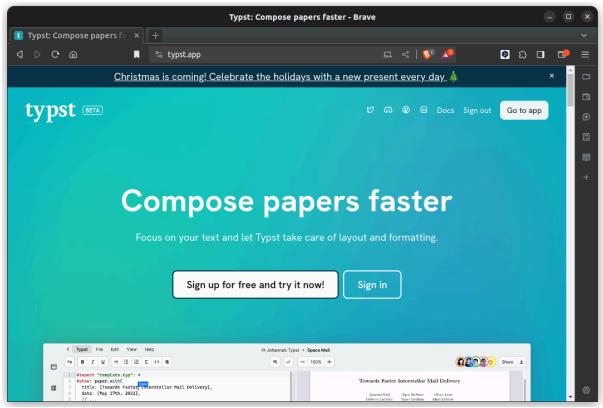

Figure 2: **Typst** app

Kindly submit your lab reports in PDF format. Before the scheduled lab period concludes, ensure you have reviewed your files with your instructor.

Figure 3: Submission portal

Before submitting your lab report, verify:

- Code builds and runs (include exact commands)
- Console output and screenshots accompany each task
- Public repository link is provided
- Document uses the **Typst** template with proper captions for figures/tables.

III. GitHub

Share your code on GitHub. It's a fantastic way to foster a supportive coding community while gaining exposure to different coding styles and techniques [2], [3], [4].

IV. Links Bundle

You may find the following links useful:

- GitHub Repository *(Figure 4)*
  github.com/a-mhamdi/nlp

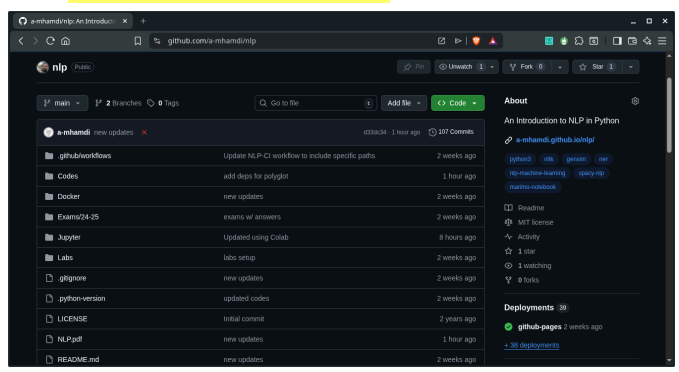

Figure 4: GitHub repository

- Docker Image *(Figure 5)*
  hub.docker.com/repository/docker/abmhamdi/nlp

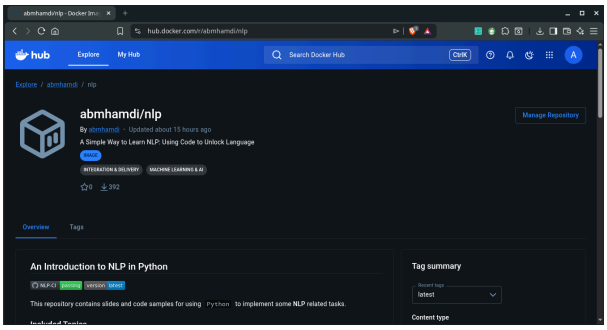

Figure 5: Docker image

V. Timeline

The following timeline is proposed to help you organize your work. It is not mandatory to follow it, but it is highly recommended to do so. The labs are designed to be completed in the order they are presented.

References

[1] T. Mailund, *Introducing Markdown and Pandoc*. Berkeley, CA: Apress L. P., 2019.

[2] S. Guthals, *GitHub*, 2nd edition. in For dummies. Hoboken, NJ: John Wiley & Sons, Inc., 2023.

[3] P. Bell and B. Beer, *Introducing GitHub*, 1. ed. Beijing: O'Reilly, 2015.

[4] M. Tsitoara, *Beginning Git and GitHub*. Berkeley, CA: Apress, 2020.